

Introduction to Validation and Theoretical Issues

[IIA – Lect.____]

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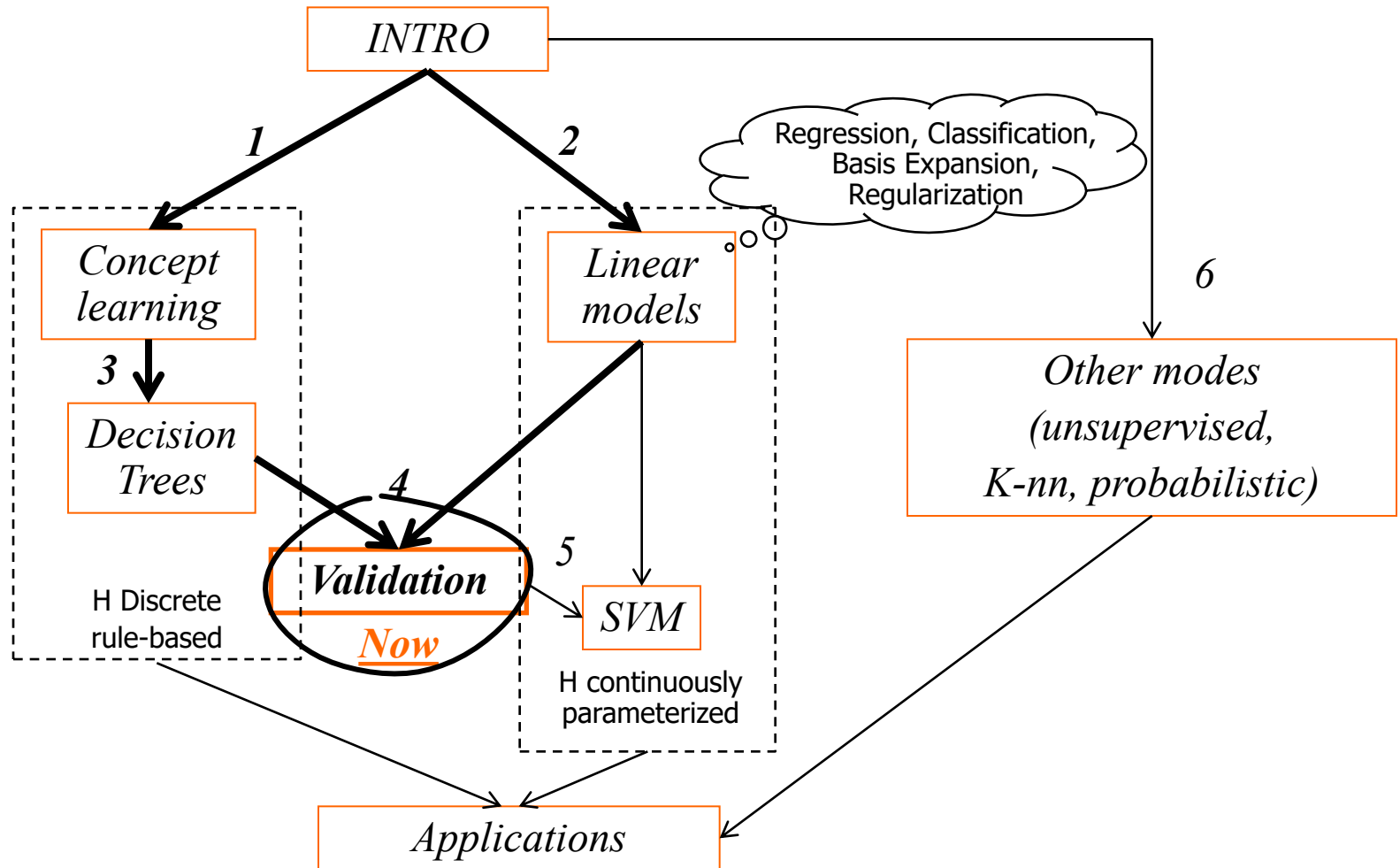


Dipartimento di Informatica
Università di Pisa - Italy

**Computational Intelligence &
Machine Learning Group**

DRAFT, please do not circulate! 2026

In the course flow



Obiettivi (overview)

Questioni fondamentali del ML:

evaluate generalization capabilities (of your hp)

- ruolo essenziale della validazione
- cenni (dell'esistenza) di fondamenti teorici (supporto al significato del ML)
- Aspetto sia teorico che pratico per un uso *consapevole* del ML
- Raccogliamo gli spunti raccolti finora solo indirettamente dedicandoci una (questa) lezione

ML issues

**Quando un modello di ML
è un buon modello?**

Usare il ML versus usare bene il ML

Machine Learning: Generalization (I)



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- *Learning*: search for a *good function* in a function space from known data

Def

- **Good** w.r.t. generalization error: it measures how accurately the model predicts over novel samples of data (low error, high accuracy and vice versa)

[Repetita from lect. 1]

Generalization (II)

- Inductive learning hypothesis
 - Any h that approximates f well on training examples will also approximate f well on new (unseen) instances x (?)
 - I.e. is it really valid? And at which extent?
- This is a **central point**, but as an objective of ML:
 - As a **theory** that explains the conditions under which this occurs
 - To guides the **choice of the “best model”** (among different models or configurations: hyperparameters, level of training, ...)
 - It **needs to be verified/measured** in applications.

Generalization (III)



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- Generalization: crucial point of ML!!! *[Repetita from lect. 1]*
- **Learning** phase (**training, fitting**): build the model from known data – *training data* (and bias)
- **Predictive** phase (**test**): apply to new examples (we take the inputs $\mathbf{x}'$ and we compute the response by the model; we compare with its target d' that the model has never seen): evaluation of the predictive hypothesis, i.e. of the **generalization capability**

Note: *performance* in ML = *predictive accuracy*
estimated by the error computed on the (Hold out) **Test Set**

Generalization (IV)



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Def

- [*repetita*] **Overfitting**: A learner overfits the data if it outputs a hypothesis $h(\cdot) \in H$ having true error ε and empirical (TR) error E , but there is another $h'(\cdot) \in H$ having $E' > E$ and $\varepsilon' < \varepsilon$

Premise: which measure?

Recap:

To evaluate typically we measure (see def. in previous lectures)

- For *classification*: MSE for the loss, accuracy or mean error rate for the outcome
 - but also precision, recall or specificity, sensitivity (accounting for False Positive, False Negative), ...
- For *regression*: MSE, Root MSE (S), Mean Absolute Error, Max Absolute Error,
 - but also statistics measures such R (correlation coefficient/index), etc.
- Of course high error $\leftrightarrow$ low accuracy (both for training, test, etc.)
 - E.g. poor fitting with high training error,
 - E.g. poor generalization with high test error, ...

Validation: Two aims

Very
important!



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Def

- **Model selection:** estimating the performance (*generalization error*) of different learning models in order to choose the best one (to generalize).

- this includes search the best *hyper-parameters* of your model (e.g. polynomial order, lambda of ridge regression, ...).

It returns a model

Def

- **Model assessment:** having chosen a final model, estimating/evaluating its prediction error/ risk (*generalization error*) on new test data (measure of the quality/performance of the ultimately chosen model).

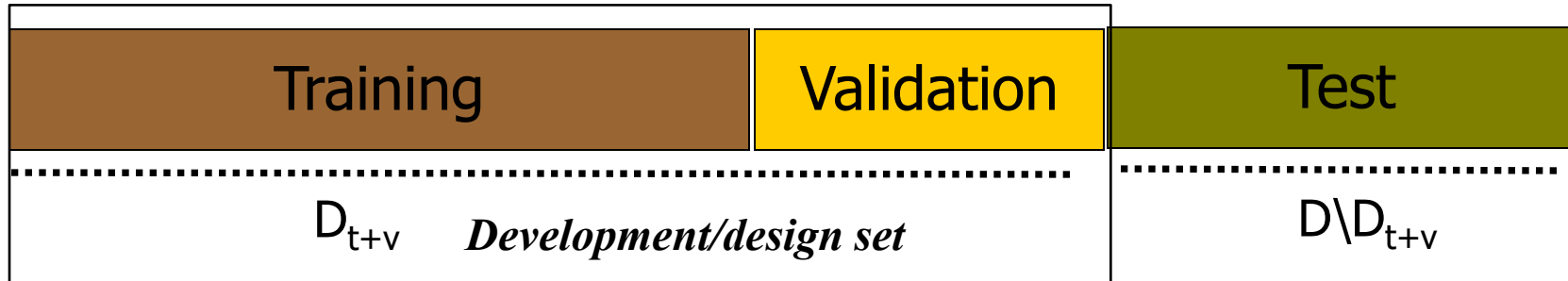
It returns an estimation

Gold rule: Keep separation between goals and use separate data sets in each phase

Defs

Hold out

- If data set size is sufficient: e.g. 50% TR, 25% VL, 25% TS **disjoint sets**



- TR**: *Training set* is used to fit [**training**]
- VL**: *Validation set* (or *selection set*) is used to select the best models (among different models and/or hyper-parameters configurations) [**model selection**]
- TR+VL sometimes are jointly called development/design set , i.e. used to build the final model
- TS**: *Test set* is used for estimation of generalization error (of the final model) [**model assessment**]

Notes:

- 1) the estimation made for model selection (on vl set) [is for model selection purpose], it is *not* a good estimation for the assessment phase/risk test.
- 2) **Test set results** cannot be used for model selection (or call it validation set)

Test or model selection?

- What if test set is used in a (repeated) design cycle?
 - We are making *model selection* and not reliable *assessment* (estimation of expected generalization error)
 - and *we wouldn't be able to do that on future examples*
 - Blind test set concept (e.g. for ML competitions)
 - Image an exam exercise: if you see the solutions it is not a test!
 - In that case, used test set error provides an overoptimistic evaluation of the true test error (→ we will see how easy is to obtain very high classification accuracy over a random task even using the test set only implicitly)

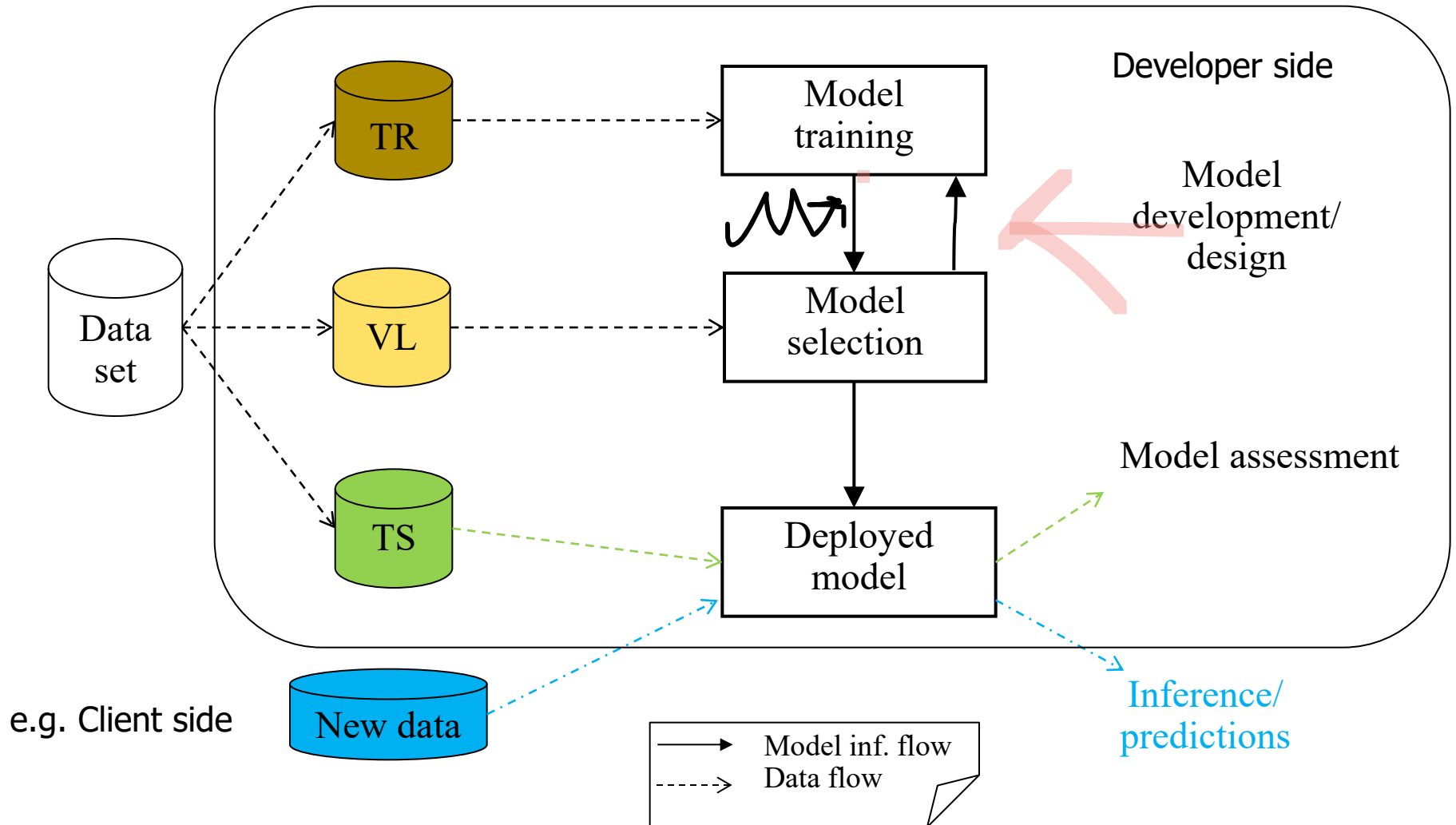
Gold rule:

Keep separation between goals and use separate sets in each phase



(**TR** for training, **VL** for model selection, **TS** for risk estimation)

TR/VL/TS by a simple schema



A simple meta-algorithm (Grid search - Hold-out)



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- Separate **TR** (training), **VL** (validation) and **TS** (test) sets
- Search **best** $h_{w,\lambda}()$ changing the model hyper-parameters λ [e.g. the polynomial order, the lambda for ridge regression]:
 - — For each different values of λ (external *grid search* loop)
 - Search **best** $h_{w,\lambda}()$ that minimize error/empirical loss (fitting the **TR set**) finding the best w parameters (internal *training* loop)
where **best** = minimum error on **TR set** [e.g. $\operatorname{argmin}_w \operatorname{Loss}(w)$ in L_2]
 - Select the best **best** $h_{w,\lambda}()$: where **best** = minimum error on the **VL set**
 - (Optional: Now it is also possible to fit $h_{w,\lambda}(x)$ on TR+VL with best λ model)
- Evaluate the final $h_{w,\lambda}(x)$ on the **TS**

This is a double (nested) cycle: Search **best** can be a "**FOR**" on a grid of values in the external cycle: for each λ value you train a model $h_{w,\lambda}$ (in the internal cycle, e.g. the gradient descent cycle) and then compute the results (accuracy) on the **VL set**. Then take the **best** value of λ i.e the model with min **VL** err or max **VL** accuracy etc.

Search on a grid (e.g. with 2 hyper-parameters)

Def

- Find *hyper-parameters* value (i.e. parameters that are not directly learnt, which are not modified by training)
- Search **best** hyper-parameter can be a “FOR” over a grid of candidate values. For each trained model $h_{w,\lambda}$ compute the results (accuracy) on the VL set. Then take the one with the minimum error or the max accuracy.

Hyper-param.	Lambda 0.1	Lambda 0.01	Lambda 0.001
Degree 1	Res1	Res4	Res7
Degree 2	Res2	Res5	Res8
Degree 4	Res3	Res6	Res9

→

E.g. “Res1” is computed on the **VL** set,
by the model with and Polynomial Degree=1
and Lambda=0.1 trained on the TR set

- Example: The best one is **Res3** → (Degree 4, lambda=0.1) is the winner
- We can automatize it!!! Parallelization is easy (independence of the trials)
- Alternatives exist to reduce the cost or to automatize the search *

Exercise

λ	TR	VL	TS
0.5	75	70	70
0.1	80	75	70
0.01	90	70	72

Accuracy (% di classificazione corretta)

- In che ordine si usano le porzioni di dati per calcolare i valori in tabella?
- Quale modello (ossia lambda) si sceglie?
- Che fenomeni si osservano?

Controesempio (separare TR, VL e TS)



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- 20-30 esempi, 1000 variabili di input random,
- *random target* 0/1
- Scelgo 1 modello con una sola variabile/feature che indovina *'per caso'* al 99% *sul dataset* ...
- e poi su qualsiasi split successivo in training, validation e test set.

Perfect result (a model with accuracy 99%)?

What is wrong?

The table for the Counterexample



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Pattern

Input variable value

	1	2	...	26	27	28	...	1000	Target
1	...	...	...	1	1	1	...	...	1
2				0	0	0			0
3				1	1	1			1
4				0	0	0			0
...				0	0	0			0
...				1	1	1			1
...				0	0	0			0
20	...	...	...	1	1	1	...	...	1
TS1				1	0	0			1
TS2				0	1	0			0
TS3				1	1	1			1
Accuracy				100%	33%	66%			

Controesempio (separare TR, VL e TS)



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Perfect result (a model with accuracy 99%)? What is wrong?
99% non è una buona stima dell' errore di test (quella corretta e' 50%)

1. Errore stimato su training o validation per model selection NON è utile per stima del rischio! Dati di TR o VL non vanno usati per scopi di test

2. Usare *tutto* il data set per feature/model selection lede la correttezza della stima (risultati biased – «Feature Selection bias»).

- Test set è stato usato implicitamente all'inizio*.
- Test deve essere separato prima, prima di qualsiasi model selection o design del modello (incluso selezione di features)

Un test set esterno fornisce invece la stima corretta del 50% (*random coin result*!).

E' la correttezza della stima che è in giudizio, non la possibilità di risolvere il task!

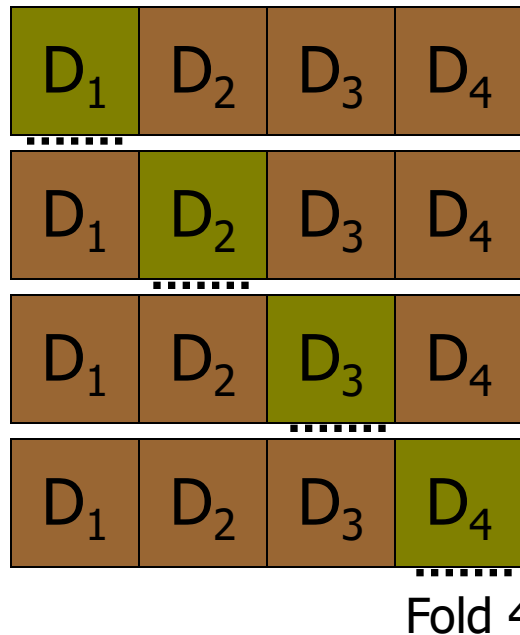
Delicato confrontando metodi diversi e usando tecniche di K-fold cross-validation che in se non garantiscono correttezza della procedura di validazione

Hold out and **K-fold** cross validation



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Hold out CV *can make insufficient use of data*



Def

K-fold Cross-Validation

- Split the data set D into k mutually exclusive subsets $D_1, D_2, \dots, D_K$
- Train the learning algorithm on $D \setminus D_i$ and test it on D_i
- Summarize averaging all the D_i results (*diagonal*)
- NOTE: This technique can be used both for the validation set or for the test set
- *It uses all the data for training and validation or testing*

Issues:

- How many folds? 3-fold, 5-fold, 10-fold, ..., 1-leave-out
- Often computationally very expensive
- Combinable with validation set, double-K-fold CV,

An **example** of model selection and assessment (with K-fold CV)



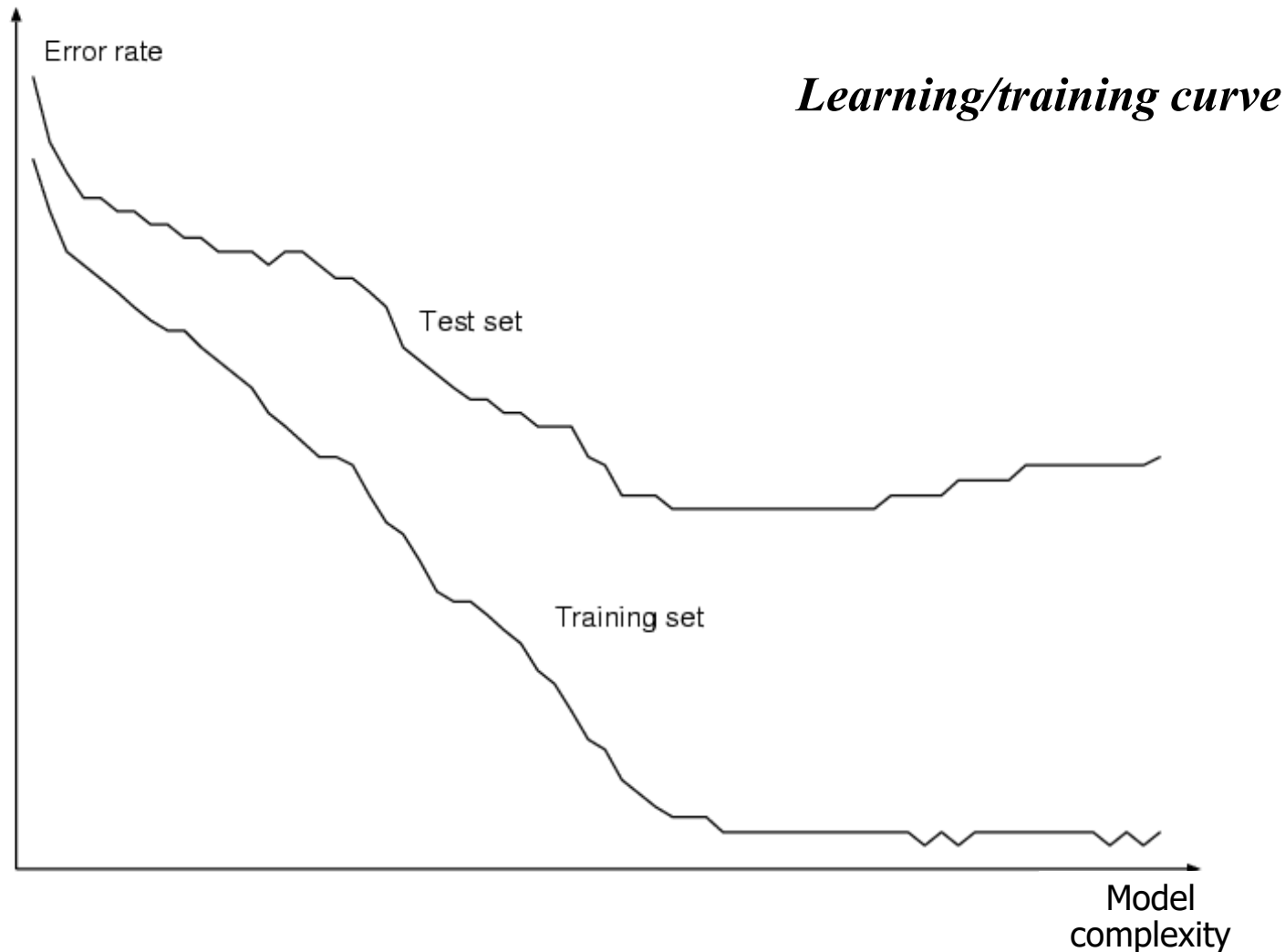
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- Split data in **TR** and **Test set** (here simple hold-out or a K-fold CV)
- **[Model selection]** Use K-fold CV (internal) over **TR** set, obtaining new **TR** e **VL** set in each split, to find best hyper-parameters of your model (e.g. polynomial order, lambda of ridge regression, ...): How?
Apply a **grid-search** with many possible values of the hyper-par.
 - i.e. for example a k-fold-CV for $\lambda = 0.1$, a k-fold-CV for $\lambda = 0.01$, ... and then take the best λ (comparing the mean errors computed over the validation sets obtained by the all the folds of each k-fold CV, ... the results on the diagonal in the previous slide)
- Train on the whole **TR** set the final model
- **[Model assessment]** Evaluate it on the external **Test set**

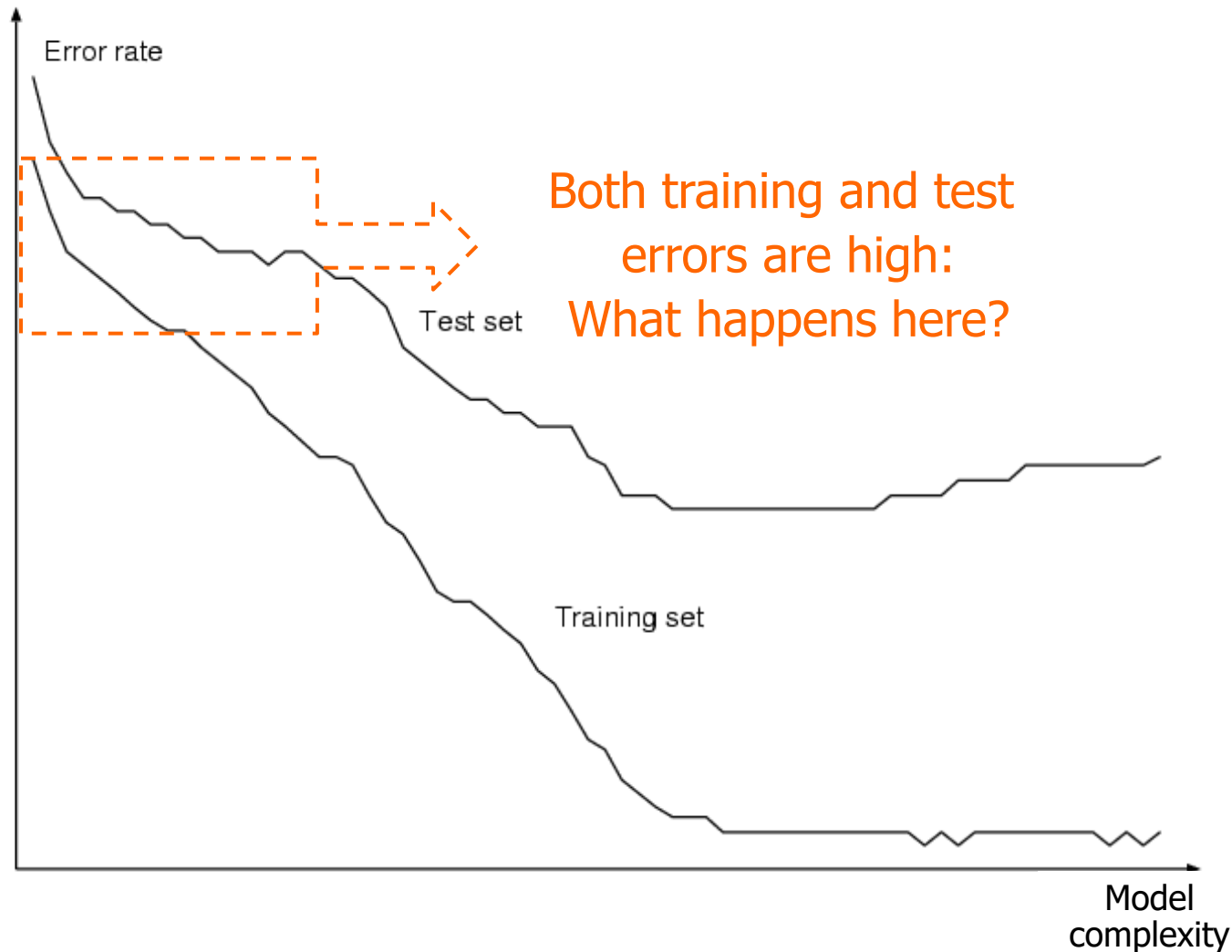
Validation: Summarizing

- **Stima Empirica:** errore calcolato/stimato su (Hold out)
 - E.g. Hold out: training, validation set and test sets
 - K-fold cross validation (resampling)
- **Teoria:** E.g. Statistical Learning Theory [Vapnik] :
 - *sotto quali condizioni (matematiche) un modello è capace di generalizzare (buona generalizzazione)?* → cenni

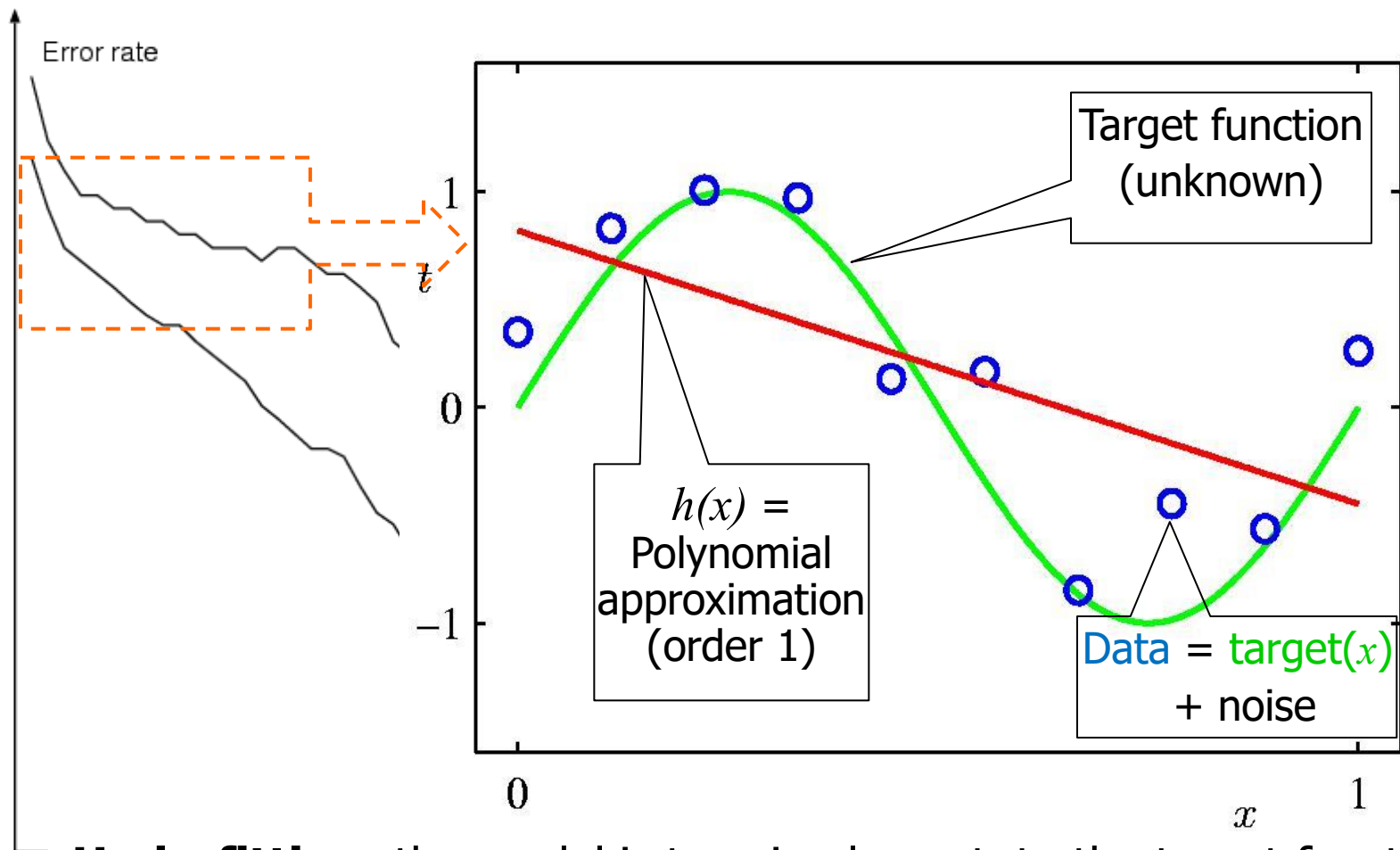
Typical behavior of learning



Typical behavior of learning

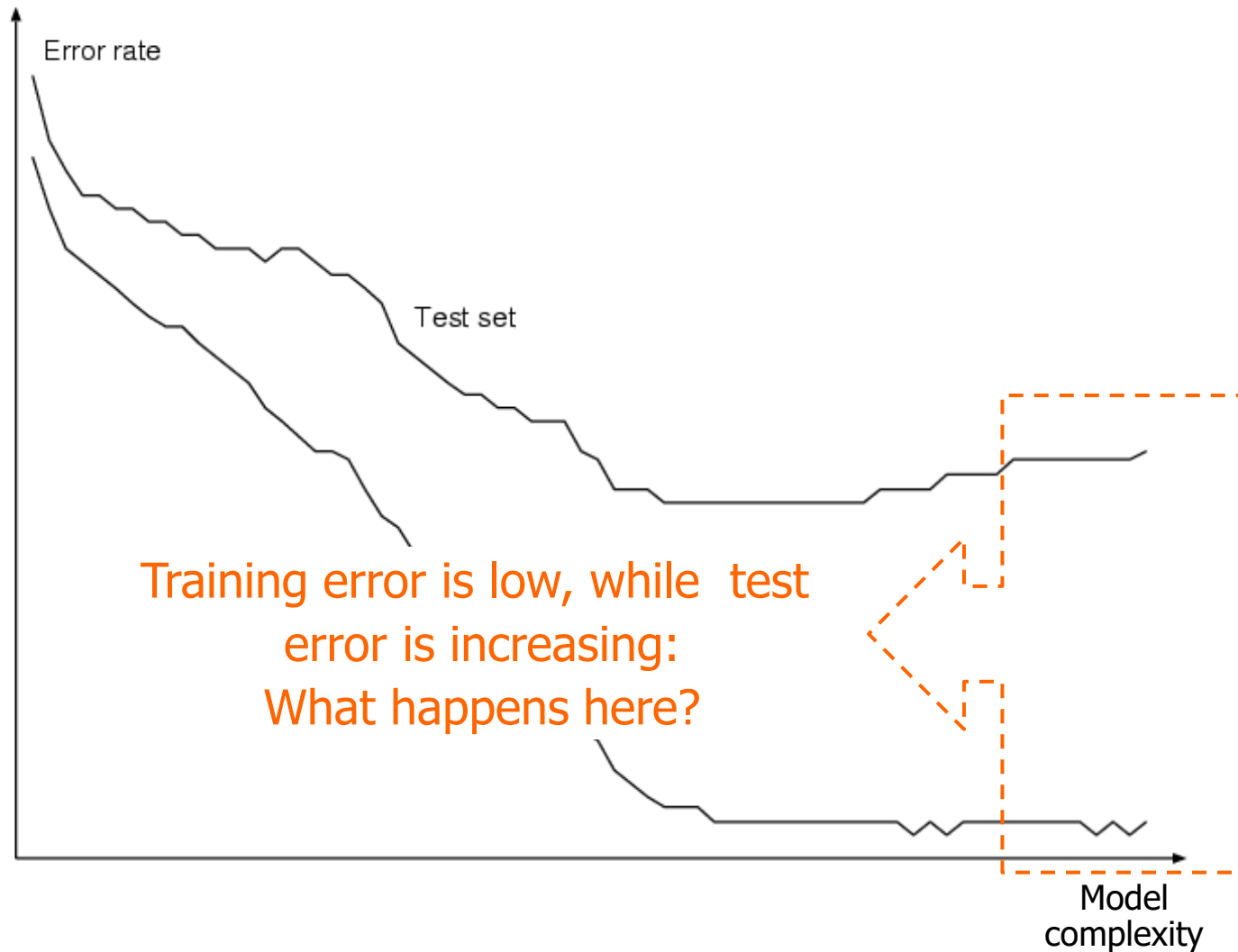


Typical behavior of learning & Polynomial Curve Fitting (I)



Underfitting: the model is too simple w.r.t. to the target function both for known data and new data

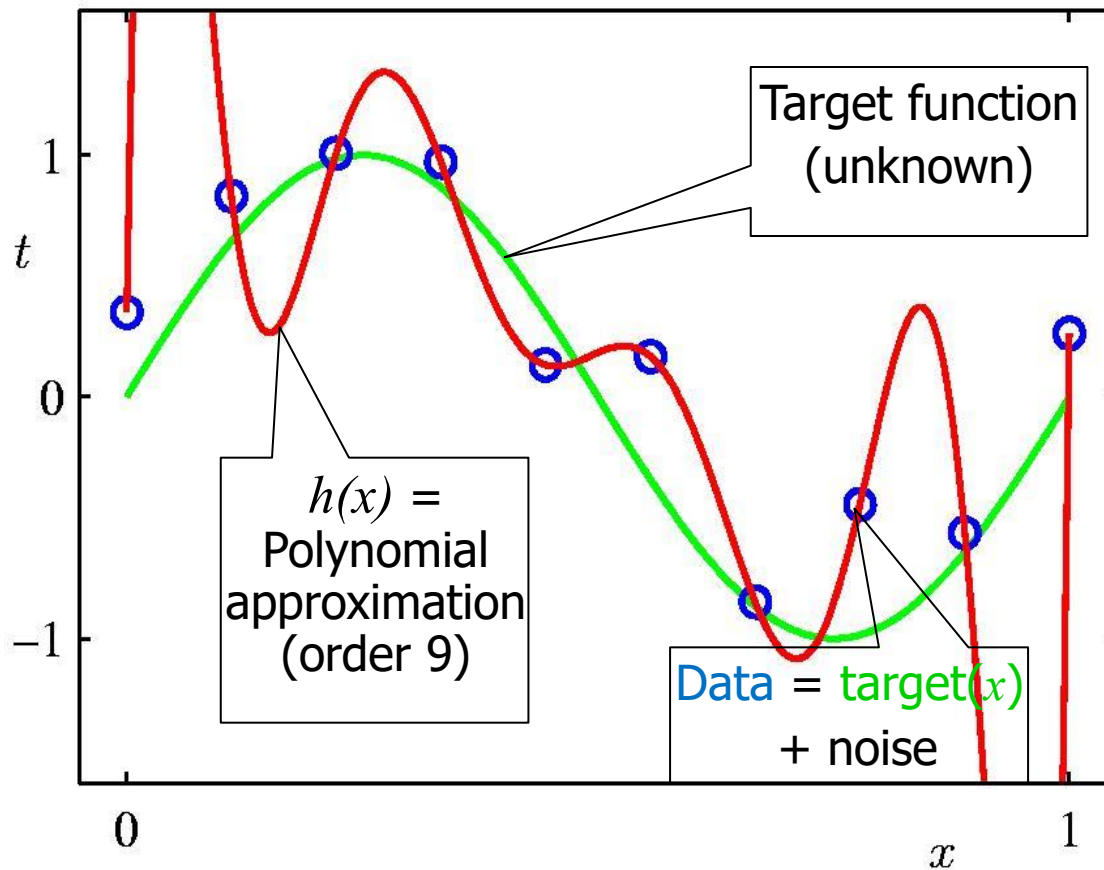
Typical behavior of learning



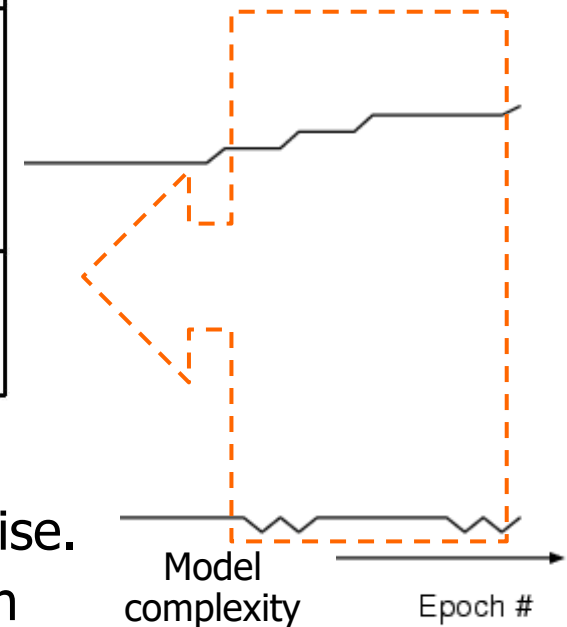
Typical behavior of learning & Polynomial Curve Fitting (II)



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Overfitting: the model is too complex: Fit the noise.
Training error is very low, test error can be high

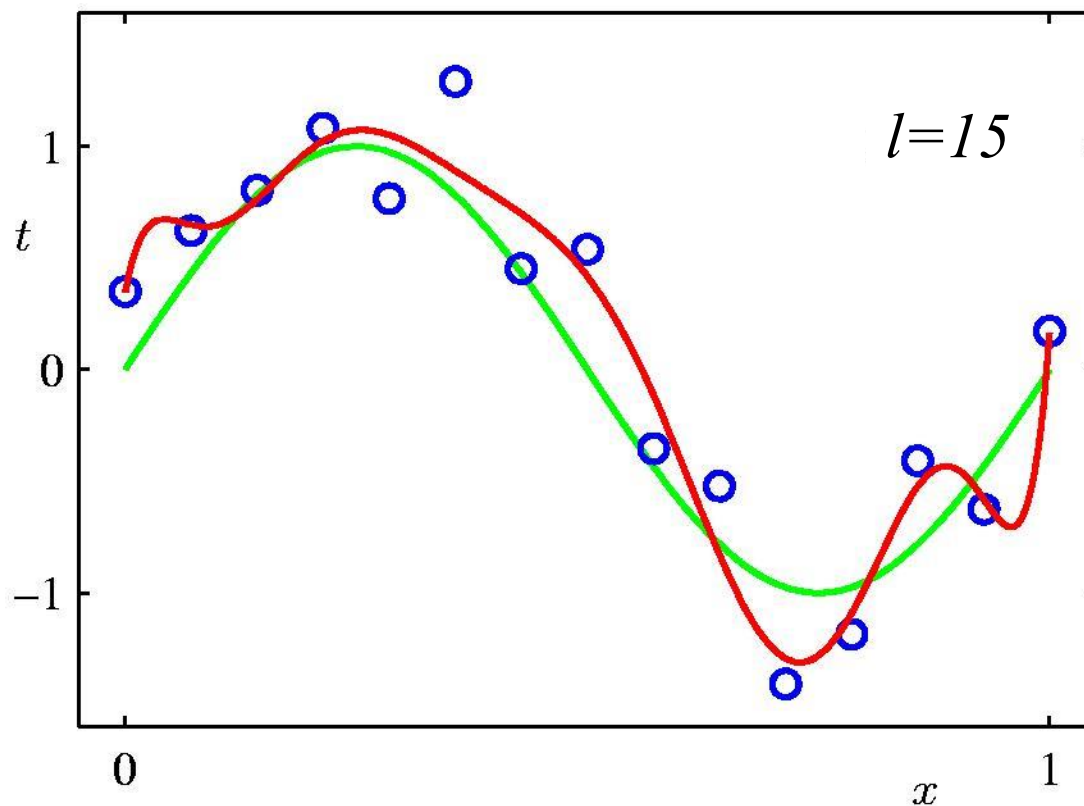


Data Set Size: $l=15$
previous was 10



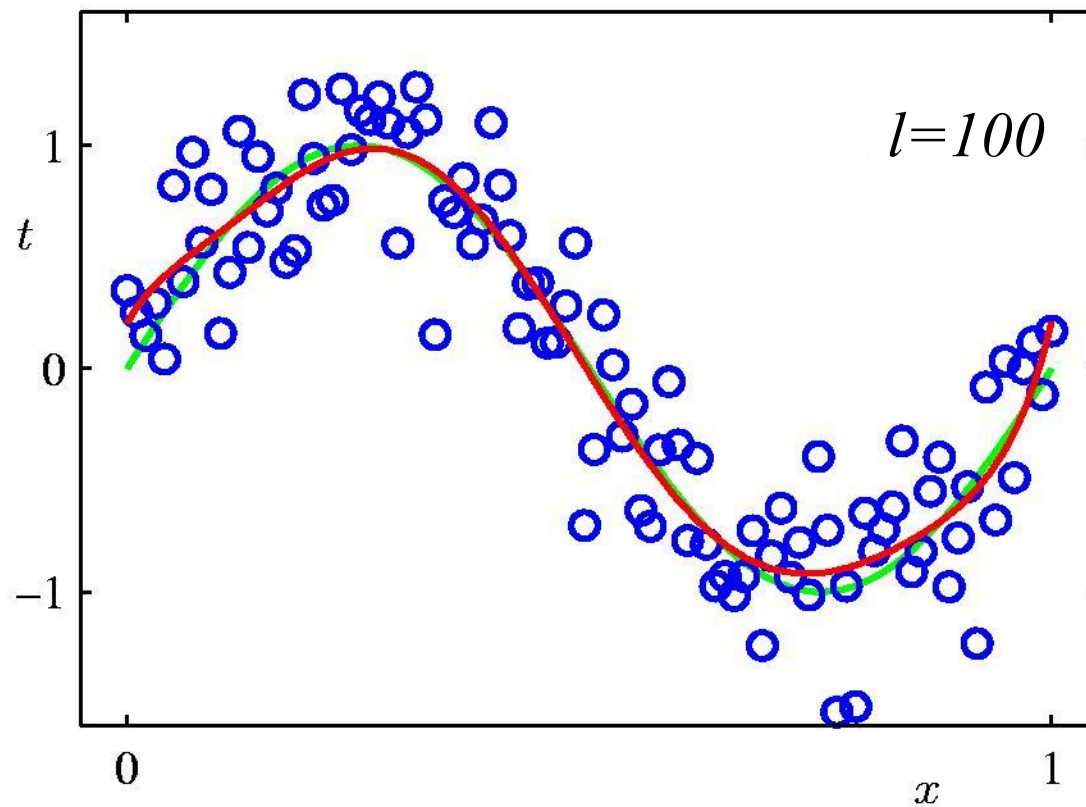
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9th Order Polynomial (changing the of number of data)



Data Set Size: $l=100$

9th Order Polynomial (even more data)



Toward SLT

Putting all together:

- The *generalization* capability (measured as a risk or test error) of a model
 - with respect to the training error
 - overfitting and underfitting zones
- 1. The role of model complexity
- 2. The role of the number of data
- Statistical Learning Theory (SLT): a general theory relating such topics

(Simplified) Formal Setting Statistical Learning Theory (SLT)

Defs

- Approximate unknown $f(\mathbf{x})$, d (or y or t) is the target ($d = \text{true } f + \text{noise}$)
- Minimize *risk function* $R = \int L(d, h(\mathbf{x})) dP(\mathbf{x}, d)$ True Error
Over all the data
- Given
 - value from teacher (d) and the probability distribution $P(\mathbf{x}, d)$
 - a loss (or cost) function, e.g. $L(h(\mathbf{x}), d) = (d - h(\mathbf{x}))^2$
- Search h in H : Min R
- But we have only the finite data set $TR = (\mathbf{x}_p, d_p), \quad p = 1 \dots l$
- To search h : minimize *empirical risk* (training error E), finding the best values for the model free parameters

$$R_{emp} = \frac{1}{l} \sum_{p=1}^l (d_p - h(\mathbf{x}_p))^2$$

- Empirical Risk Minimization (ERM) Inductive Principle
- *Can we use R_{emp} to approximate R ?*

Vapnik-Chervonenkis-dim and SLT: a general theory (I)



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Def

Given the $VC\text{-dim}$ (VC), a measure *complexity* of H (*flexibility to fit data*) (e.g. Num. of parameters for linear models/polynomials)

Repetita: Can we use R_{emp} to approximate R ?

Very
important!

Def

$VC\text{-bounds in the form:}$ it holds with probability $1-\delta$ that

guaranteed risk

$$R \leq \underbrace{R_{emp}} + \underbrace{\varepsilon(1/l, VC, 1/\delta)}_{VC\text{-confidence}}$$

- First (basic) explanation:
 - ε is a function that grows with VC ($VC\text{-dim}$), that decreases with (higher) l and δ (i.e. l and δ are in the denominator).
 - We know that R_{emp} decrease using complex models (with high $VC\text{-dim}$) (e.g. the polynomial degree in the example)
 - δ is the confidence, it rules the probability that the bound holds (e.g. low delta 0.01, it holds with probability 0.99)
- Now we can see how it can “explain” the *underfitting* and *overfitting* and the aspects that control them.

Vapnik-Chervonenkis-dim and SLT: a general theory (II)



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Comments:

Very important!

- *VC-bounds in the form:* it holds with probability $1-\delta$ that

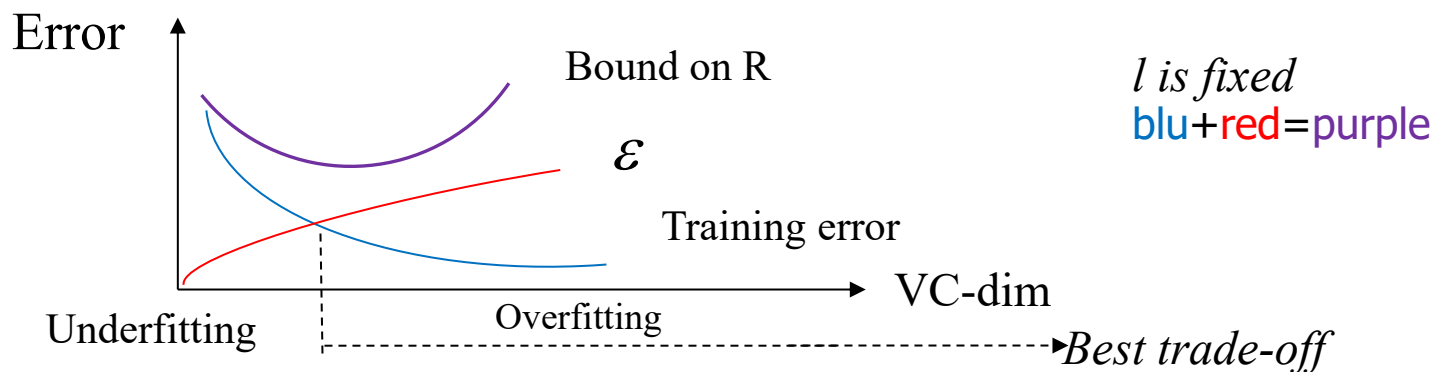
Def

guaranteed risk $R \leq \underbrace{R_{emp}}_{\text{VC-confidence}} + \underbrace{\varepsilon(1/l, VC, 1/\delta)}_{\text{VC-confidence}}$

Intuition:

- Higher l (data) $\rightarrow$ lower *VC-confidence* and R_{emp} closer to R
- Too simple model (low VC-dim) can be not suff. due to high R_{emp} (underfitting)
- Higher VC-dim (fix l) $\rightarrow$ lower R_{emp} but *VC-conf.* and hence R may increase (overfitting)

Def • *Structural risk minimization:* minimize the bound !



- Concept of control of the model complexity (flexibility):
trade-off between TR accuracy (fitting) and model complexity (VC-dim)

Discussion

Complexity control



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- **Statistical Learning Theory (SLT)**

- Permette inquadramento formale del problema della generalizzazione e (underfitting/)/overfitting, fornendone limitazioni superiori analitiche e quantitative al rischio R di predizione su tutti i dati, indipendentemente dal tipo di learning algorithm o dettagli del modello.
- Il ML è ben fondato:
 - Il rischio del learning (e l'errore di generalizzazione) può essere analiticamente limitato, e solo pochi concetti sono fondamentali !
 - Si può trovare una buona approssimazione della f target da esempi, pur di avere un buon numero di dati e una adeguata complessità del modello (misurabile formalmente con la VC-dim)
- Porta a nuovi modelli (SVM) (e altri metodi che direttamente considerino il controllo della complessità nella costruzione del modello)
- Fonda uno dei principi induttivi sul *controllo della complessità*

Domande aperte:

- Quali (altri) principi vi sono per fondare il controllo della complessità e come operare in pratica?
 - Come misurare la complessità (flessibilità per il fitting)?
 - Come trovare il bilanciamento ottimo tra fitting e complessità ?

Some Examples for Complexity Control



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- Linear models (LM):
 - Complexity seems* related to number of free parameters w : input dimension / dim. of the basis expansion (e.g. polynomial degree)
 - Lambda parameter for the regularized version (using the model selection/validation techniques to find the proper value of lambda)
- Decision trees (DT): number of nodes (e.g. control by early stop, pruning)
- We will also see: direct approach to the complexity optimization through the SVM model
- **Exercise**: relate the complexity control to the approaches used in the different models, explaining the *underfitting* and the *overfitting* from the point of view of the SLT upper bound on R : e.g. how to explain the role of the hyper-parameters lambda in Linear models or # of nodes in DT etc. in terms of SLT?

Conclusioni

- ML models flexibility →
 - Use the power of ML without control is a way to produce *illusory results*
 - Control the tradeoff between model fitting and complexity
 - Fundamental role of validation approaches (for model selection and estimations)
- Il ML è ben fondato teoricamente
 - Domande fondamentali tramite Statistical Learning Theory (#ML)
 - ed altre (e.g. PAC- probably approximately correct learning con cenni in AIMA cap. 18.5)

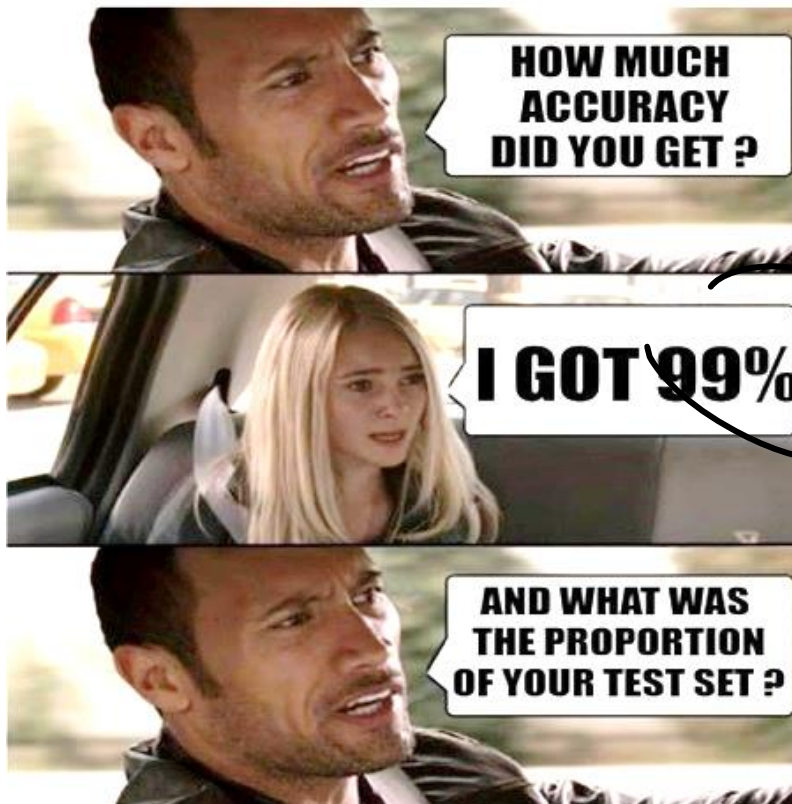
Bibliography

- AIMA , ed .3: chap 18.4
- AIMA ed. 4: 19.4.1 e 19.4.4 (cenni alla sola model selection)
- Further readings (not mandatory!):
- Every good ML book:
 - see the bibliography of the SVM lecture

For fun

- Can I have just a look to the test set?

See <https://youtu.be/XvOsh15hLIIs>



By Sepe-Dukic past ML students



* It can hold also for the validation set used as test set ;-)

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